EV Technology Course — Module 12: Testing, Validation and Homologation

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Before an EV can be sold, every component and the complete vehicle must be proven to work, survive and be safe, and the vehicle must be type-approved (homologated) by the government's testing agency. In India that means certification under the Central Motor Vehicles Rules (CMVR) at agencies such as ARAI, ICAT or GARC, against Automotive Industry Standards (AIS). This module explains how that proof is built.

1. Learning objectives

- Explain verification, validation and homologation and how they differ.
- Describe the V-model of automotive development and testing levels (MIL, SIL, HIL, vehicle).
- List key EV-specific tests: battery safety, EMC, electrical safety, range and energy consumption, durability and environmental.
- Know the main Indian standards and agencies and their global equivalents.
- Calculate test durations and reliability sample sizes.
- Understand how test plans drive product development cost and timing.

2. Fundamentals

- **Verification:** did we build it right? (Does the part meet its specification?)
- **Validation:** did we build the right thing? (Does the vehicle meet customer and real-world needs?)
- **Homologation / type approval:** does the vehicle meet the law? Done by an authorised agency and leads to a certificate allowing registration and sale.
- **Conformity of production (CoP):** periodic checks that vehicles coming off the line still match the approved type.
- Tests happen at every level: material, component, subsystem, system and vehicle.

3. Detailed technical explanation

The V-model

- Left side: requirements → system design → subsystem design → component design.

- Bottom: build hardware and code software.
- Right side: component tests → subsystem integration → system integration → vehicle validation, each checking the matching design level on the left.

Test levels for software and controls

Level	What is real	Use
MIL (model-in-the-loop)	Nothing physical; models only	Early algorithm design
SIL (software-in-the-loop)	Real code on a PC	Code verification, regression
HIL (hardware-in-the-loop)	Real ECU with simulated vehicle	Faults, edge cases, automation
Vehicle-in-the-loop / road	Real vehicle	Calibration and final validation

EV-specific tests

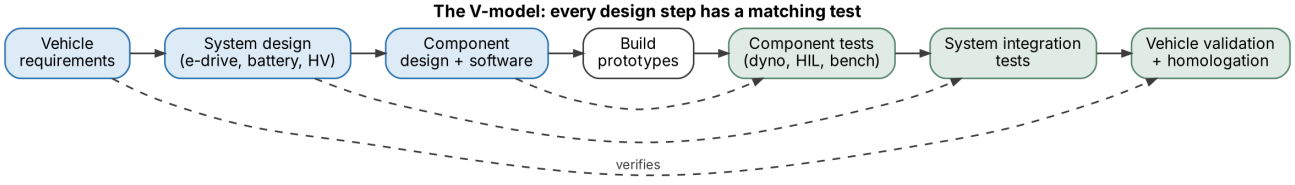
Area	Typical tests	Standards (India / global)
Battery safety	Vibration, mechanical shock, crush, fire resistance, external short circuit, overcharge, over-discharge, thermal shock, thermal propagation, IP rating	AIS-038 Rev 2 (four-wheelers, buses, trucks), AIS-156 (two- and three-wheelers) / UN R100, GB 38031
Electrical safety	Isolation resistance, protection against direct and indirect contact, IPXXB/IPXXD, HV markings, post-crash HV safety	AIS-038 / UN R100, ISO 6469
EMC	Radiated and conducted emissions, immunity, during charging too	AIS-004 Part 3 / UN R10, CISPR 25
Energy and range	Energy consumption and range on drive cycle (MIDC in India)	AIS-039 / WLTP, EPA
Power	Net power and 30-minute maximum power of the e-motor	AIS-041 / UN R85
Charging	Connector, communication and safety	AIS-138, IS 17017 / IEC 61851, IEC 62196

Area	Typical tests	Standards (India / global)
Crash	Frontal and side crash, pedestrian protection, Bharat NCAP star ratings	AIS-098, AIS-099, AIS-197 (Bharat NCAP) / UN R94, R95
Functional safety	Hazard analysis, ASIL, safety cases	ISO 26262
Environmental and durability	Thermal cycling, humidity, salt spray, dust, water wading, vibration, road durability	IEC 60068, OEM specs

Development test phases (typical)

- Mule vehicles (old body, new powertrain) → prototype builds (often called VP or P1/P2) → pre-production (PP/PPAP-level parts) → SOP.
- Each phase has a test plan (DVP&R: design verification plan and report) with test items, sample sizes, responsibilities and results.

4. Diagram: the V-model



5. Step-by-step: from test plan to certificate

1. Derive requirements from regulations, customer targets and internal standards.
2. Build the DVP&R: every requirement linked to a test, sample size and acceptance criteria.
3. Run component and subsystem tests (bench, HIL, environmental chambers).
4. Run vehicle tests: durability, hot and cold climate, high altitude, EMC, range, NVH, crash.
5. Fix failures through root-cause analysis and design changes (ECNs), then retest.
6. Submit to the testing agency with technical documents; agency witnesses or conducts tests.
7. Receive the type-approval certificate; plan CoP checks during production.

6. Engineering equations

Test duration:

$$t = \frac{\text{distance}}{\text{average speed} \times \text{hours per day}}$$

Success-run (zero-failure) sample size for reliability R at confidence C:

$$n = \frac{\ln(1 - C)}{\ln R}$$

Acceleration factor for compressed durability:

$$AF = \frac{\text{customer-equivalent damage}}{\text{test distance}}$$

7. Worked examples

Example 1: road durability. A 250,000 km life target at 60 km/h average, 20 hours per day, takes 250,000 ÷ 1,200 = **208 days**. On a rough-road track with an acceleration factor of 10, it needs 25,000 km = **21 days**.

Example 2: reliability sample size. To demonstrate 90% reliability with 90% confidence and zero failures: n = ln(0.1) ÷ ln(0.9) = **22 samples**. For 95% reliability, 45; for 99%, 230. This is why high-reliability claims need either many samples or longer, accelerated testing.

Example 3: thermal cycling. 1,000 cycles of 2 hours each = 2,000 hours = **83 days** of chamber time, which drives the project timeline and chamber capacity planning.

8. Real-world examples

- **Bharat NCAP** was launched in August 2023, with testing from October 2023 and first results in December 2023; EVs such as the Tata Punch.ev, Nexon EV and Mahindra BE 6 / XEV 9e have received 5-star ratings.
- **AIS-156 and AIS-038 Rev 2** battery safety amendments were tightened after e-scooter fires in 2022, adding thermal propagation and cell-level requirements.
- **PM E-DRIVE and earlier FAME** subsidies require testing agencies to certify performance and local value addition.
- Global OEMs test in extreme climates: hot tests in the Middle East and Rajasthan, cold tests in Scandinavia and Ladakh.

9. Manufacturing applications

- **End-of-line (EOL) testing** in the plant: HV isolation, HVIL, charging check, motor spin, ADAS calibration, software flashing, roller-dynamometer run, water leak test.
- **PPAP and APQP:** supplier parts must pass validation (PV) tests before production approval.
- **CoP audits:** sample vehicles from production re-tested to confirm compliance.
- **Test cost:** prototype vehicles, chamber time, agency fees and crash cars are a large share of development cost; costing engineers must budget DVP&R costs early.

10. Common mistakes and misconceptions

- **"Passing homologation means the product is good."** It only shows legal compliance; customer durability is separate.
- **"Simulation replaces testing."** It reduces tests but cannot replace physical validation and certification.
- **"One successful test proves reliability."** Statistically you need sample sizes and confidence levels.
- **"Testing is the last step."** In the V-model, test planning starts with requirements.

11. Practical case study: an EMC failure late in development

Problem. At pre-production, the vehicle fails radiated emission limits in the FM band during DC fast charging.

Root cause. The inverter and DC-DC converter switching harmonics couple into an unshielded HV cable routed near the antenna amplifier.

Fix. Shielded HV cable with 360° connector termination, ferrite on the LV harness, revised cable routing and changed switching frequency. Two months of delay and ₹ cost of new harness tooling.

Lesson. EMC must be designed in from the architecture stage with early pre-compliance tests.

12. Revision notes

- Verification = right build; validation = right product; homologation = legal approval.
- V-model: each design level has a matching test level.
- India: CMVR, AIS standards, ARAI/ICAT/GARC; battery safety AIS-038 Rev 2 / AIS-156; EMC AIS-004 Part 3; range AIS-039.

- Zero-failure sample size n = ln(1−C) ÷ ln R.

13. Glossary

Term	Meaning
CMVR	Central Motor Vehicles Rules
AIS	Automotive Industry Standard (India)
ARAI / ICAT / GARC	Indian testing and certification agencies
DVP&R	Design verification plan and report
HIL	Hardware-in-the-loop testing
EMC	Electromagnetic compatibility
CoP	Conformity of production
EOL	End-of-line testing

14. Knowledge check

Conceptual

1. Difference between verification and validation?
2. Why is HIL useful for BMS testing?
3. Why were battery safety standards for two-wheelers tightened in India?
4. What is conformity of production?
5. Why must EMC be considered during charging?

Numerical

1. Zero-failure samples needed for 95% reliability at 90% confidence?
2. 150,000 km durability at 50 km/h, 16 h/day. Days?
3. A test track has an acceleration factor of 8. Track distance for a 240,000 km target?

Application

1. Draft the main headings of a DVP&R for an EV traction inverter.
2. As a costing engineer, list the main cost buckets in validation for a new EV.

15. Answer key

Conceptual

1. Verification checks against the specification; validation checks against real customer needs and use.
2. It simulates cells, faults and extreme conditions safely and repeatably, without damaging real packs.
3. A series of e-scooter fires in 2022 showed gaps; amendments added thermal propagation, cell testing and BMS requirements.
4. Periodic checks of production vehicles to confirm they match the approved type.
5. High-power switching in the charger and vehicle can emit interference and must also be immune to grid disturbances.

Numerical

1. $\ln(0.1) \div \ln(0.95) = 44.9 \rightarrow$ **45 samples**.
2. $150{,}000 \div (50 \times 16) =$ **187.5 days**.
3. $240{,}000 \div 8 =$ **30,000 km**.

Application

1. Electrical performance (efficiency, torque accuracy), thermal (coolant limits, derating), EMC, environmental (temperature cycling, humidity, salt, vibration, shock), insulation and HV safety, functional safety (fault injection), durability (power cycling of modules), software verification, sample sizes and acceptance criteria.
2. Prototype vehicles and parts, test facility and chamber time, agency certification fees, crash vehicles, climate test trips, HIL rigs and software tools, engineering manpower, re-tests after failures.

16. Further learning

- ARAI and MoRTH websites for AIS standards (many are free to download).
- UNECE regulations R10, R100, R94/R95.
- ISO 26262 overview from TUV or SGS.
- AIAG APQP and PPAP manuals.

Next: Module 13, EV manufacturing and cost engineering.